Computer Project 1: Explicit Burgers Equation Solver
Handed out 19th September 2026; Due by March 26th, 2026

Write a computer program to solve the inviscid Burgers equation (1) using explicit, finite-volume techniques.

$$\frac{\partial u}{\partial t} + \frac{\partial}{\partial x}\left(u^2\right) = 0 \tag{1}$$

The extents of your 1-D domain are from $x = -1$ to $x = 1$. Use 21 grid points and a time-step of 0.01 s (or less if this does not work).

- $u = 1$, $x \leq 0$; $u = 0$, $x > 0$;
- $u = 0$, $x \leq 0$; $u = 1$, $x > 0$;
- $u = -1$, $x \leq 0$; $u = 1$, $x > 0$;
- $u = 1$, $x \leq 0$; $u = -1$, $x > 0$;
- $u = 1$, $x \leq -0.5$; $u = 0$, $x \geq 0.5$; u is continuous and varies linearly between $-0.5 \leq x \leq 0.5$

Assume that the ghost-cell values are same as the immediate internal neighbours.

Please use the following three methods: a flux-splitting method, a flux-difference splitting method and Godunov's method (discussed in class) to construct the flux at the interfaces. Your output should include u vs. x plots, at $t =$

- 0.1 s
- 0.5 s

Please also include the exact solution in the plots to compare with your numerical solution.

You need to present your report in a technical paper format. In your report

- In your title page please write your names and roll numbers. At the end of your report include the code as an Appendix item.
- Provide some description/explanation for any plot you present in the write up. You should not just post a figure without commenting on it.
- Please submit a pdf copy of your report with the following naming convention: AS5460-Group-xx-Project-1.pdf where xx refers to your group number.
- **I would expect each group to write their own code. No sharing of code is acceptable between groups and absolutely no collaboration on the reports is permitted. This can result in getting a U grade in this course.**